
Channel-Set Drift in Long-Reasoning W4A16 LLMs: Mechanisms and a Budget-Tuned Remedy

Anonymous authors

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Abstract

Static per-channel protection is attractive for W4A16 reasoning inference, but it assumes that high-magnitude activation channels remain useful protection targets throughout decode. We measure decode-time channel-set drift across four reasoning LLMs and find strict top-1% set-leaving of 53–67% by horizons up to 20K tokens. The effect appears at the measured block-output surface in hybrid Mamba-2, parallel-hybrid, and pure-Transformer models, extending DecDEC’s short-horizon Transformer observation and challenging the protected-set persistence that Quamba2-style static ordering would require at this measured surface. We then test eight channel-set protection families with matched-cost and random controls. Hard position or scale switches are actively harmful, top-1% smoothing and cross-tensor coupling have signal but insufficient recovery, and reactive DecDEC-style selection does not recover the W4A16 gap. Budget-tuned EMA protection is the strongest channel-set remedy: on Nemotron-3, top-10 protection recovers 0.82 of the BF16-vs-static gap and beats static top-10 by 0.22; on Granite-Small, top-5 recovers 0.45 with a wide CI. A ParoQuant-style rotation baseline recovers 0.75 on Granite-Small, scoping our negative results to channel-set protection rather than W4A16 PTQ generally. KL trajectories grow sublinearly with AR decays 0.44–0.51, weakening compound-error explanations in the measured regime.

1 Introduction

Long-reasoning models can spend thousands of decode steps in a single answer, so calibration-time quantization decisions must survive within-trace distribution shift. Static protected-channel maps are especially brittle if the channels that dominate early activation magnitude do not remain dominant later. This paper studies that assumption for W4A16 post-training quantization (PTQ): 4-bit weights, 16-bit activations, and a small protected activation-channel set kept at higher precision.

The assumption matters for recent state-space quantization. Quamba2 argues that Mamba-family activations exhibit channel-order preservation and activation persistence, properties that support static channel ordering and clustering (Chiang et al., 2025). We test the corresponding protected-set assumption at long reasoning horizons. Across Granite-4.0-H-Small, Nemotron-3-Nano, DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1, 53–67% of the top-1% channels at the calibration position strictly leave the later top-1% set. This is not merely within-set rank shuffling; it is membership change in the set a static policy would protect.

The systems motivation is narrow but practical. Reasoning workloads are expensive, and inference-cost forecasts expect large pressure to reduce per-token cost even as reasoning demand grows (Bort, 2025; Gartner, 2026). Recent W4A16 systems for larger standard Transformers can recover substantial quality, including ParoQuant’s rotation-based reasoning results (Liang et al., 2025). Our focus is different: small reasoning models, hybrid Mamba-2 or parallel-hybrid architectures, pure PTQ rather than distillation, and channel-set protection under long decode.

Our contributions are fourfold.

- Empirical drift. We provide a four-model cross-architecture measurement of decode-time channel-set drift up to 20K tokens, with strict set-leaving of 53–67% across hybrid Mamba-2, parallel-hybrid, and pure-Transformer models.
- Prior-art differentiation. We extend DecDEC’s short-horizon Transformer evidence to the long-reasoning regime and show that Quamba2-style protected-set persistence does not hold at the measured block-output surface for the hybrid models tested here.
- Mechanism tests. We evaluate eight channel-set protection families with random and matched-cost controls, yielding a three-mechanism account: boundary discontinuities are harmful, smooth top-1% tracking is insufficient, and protected budget is load-bearing; KL accumulation does not support runaway compound error in the measured Granite packet.
- Partial remedy and scope. Budget-tuned EMA protection succeeds on Nemotron at top-10 and shows partial Granite signal, while ParoQuant recovers 0.75 on Granite via rotation. Thus the negative result is scoped to channel-set protection, not all W4A16 PTQ.

2 Related Work

Mamba and state-space quantization. Vision-Mamba quantization already reports dynamic activation outliers: QMamba introduces temporal grouping for hidden states (Li et al., 2025), and OuroMamba uses dynamic outlier detection during inference (Ramachandran et al., 2025). Language-Mamba work often assumes more stable structure. Mamba-PTQ identifies recurrent outlier channels (Pierro and Abreu, 2024); Quamba and Quamba2 use static or calibration-time structure for Mamba-family PTQ (Chiang et al., 2024; 2025); MambaQuant uses variance-aligned rotations (Xu et al., 2025). Our measurement asks whether the static protected-set assumption holds in long reasoning traces.¹

LLM quantization and rotations. SmoothQuant, AWQ, QuaRot, KVQuant, and BlockDialect protect or transform activation structure using calibration-time statistics, rotations, or adaptive formats (Xiao et al., 2023; Lin et al., 2023; Ashkboos et al., 2024; Hooper et al., 2024; Jang and Tambe, 2025). These systems are not contradicted by our results. The implication is narrower: if the protected channel set changes during long decode, static channel maps need validation on the specific architecture and horizon. ParoQuant is particularly important because it reports strong reasoning-task accuracy under low-bit PTQ using pairwise rotations and channel-wise scaling (Liang et al., 2025); our Granite ParoQuant-style baseline tests whether a rotation mechanism, rather than channel-set protection, can recover the endpoint measured here.

Dynamic and long-context policies. DecDEC dynamically identifies salient activation channels at each decode step and reports low recall for static outlier-channel identification in short Transformer decode (Park et al., 2025). Relative to that setting, our extension is longer reasoning horizons, strict set-membership drift accounting, hybrid Mamba/parallel-hybrid coverage, and reasoning-trace intervention controls; the pure-Transformer result is a falsification/control surface. KV-cache methods such as KIVI, PM-KVQ, AttentionPredictor, ChanMix, and MixKVQ adapt or compress cached key/value tensors (Liu et al., 2024; 2025; Yang et al., 2025; Liao and Wen, 2026; Zhang et al., 2025); our interventions target linear-layer activation channels.²

Error accumulation. SLQ shows that quantized LLMs can preserve task-level and next-token distribution fidelity under sufficiently careful bit allocation (Helcig et al., 2026). QEP

¹We use decode-time channel-set drift for the measured phenomenon. SmoothQuant’s “migration” is activation-to-weight scale transfer (Xiao et al., 2023); MoBiQuant uses related language for token-set shifts across bit-widths (Wang et al., 2026).

²“Hot channels” in HCP/CHON refers to persistently extreme channels during NVFP4 pretraining, which differs from our W4A16 inference-time setting. We refer to arXiv:2603.19296 as test-time TTQ to avoid conflict with classical Trained Ternary Quantization.

emphasizes cross-layer quantization-error growth, LAQuant emphasizes layer-wise look-ahead losses and calibration/deployment alignment for reasoning models, and Activation Sensitivity formalizes channel-wise perturbation impact on loss and downstream propagation (Arai and Ichikawa, 2025; Choi et al., 2026; Xu, 2026). These results motivate testing whether our more aggressive W4A16 endpoint exhibits decode-position error growth rather than assuming it.

3 Method

For each trace, layer, and measured decode position, we record the vector of per-channel activation magnitudes. Let $S_{\ell,p}(t)$ be the top-1% channel set for layer ℓ and prompt p at position t . The strict set-leaving rate at position t is

$$1 - \frac{|S_{\ell,p}(100) \cap S_{\ell,p}(t)|}{|S_{\ell,p}(100)|}, \quad (1)$$

averaged over prompts and layers. Within-set shuffling is reported separately for channels that remain in the top-1% set but move substantially in rank.

We score interventions using recovery of the BF16-vs-static W4A16 gap:

$$R = \frac{L_{\text{static}} - L_{\text{method}}}{L_{\text{static}} - L_{\text{BF16}}}. \quad (2)$$

Recovery may be negative when a method is worse than static protection and may exceed one when a method beats BF16 on the scored slice. We therefore report unclipped medians and bootstrap CIs. Traces with no recoverable static gap are tracked explicitly and excluded from ratio computation for method CIs; their frequency is part of the trace-heterogeneity result.

We test eight channel-set method classes: static migration-aware unions, hard position-conditioned protected sets, hard position-binned scale tables, EMA-smoothed top-magnitude protection (M11), cross-tensor activation+key coupling (M18), DecDEC-style reactive top-1% selection, budget-tuned EMA protection (M11b), and stable-core protection (M26). Each method is compared to static baselines plus random or matched-budget controls when applicable. ParoQuant is evaluated as an algorithmic rotation baseline, not as a channel-set method.

4 Experiments

Models and traces. The measurement suite covers Granite-4.0-H-Small, Nemotron-3-Nano, DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1. Intervention experiments use 12 deterministic long-reasoning traces unless otherwise noted; the original drift measurement uses 24 Granite-Small traces. Decode positions are measured on the grid $\{100, 500, 1000, 5000, 10000, 20000\}$ when the model packet supports 20K tokens. Dense KL analysis uses a Granite-Small packet sampled every 100 positions to 10K.

Controls. Static top-1% protection is the baseline. Position- and scale-switching methods use random-bin controls. EMA and reactive methods use random-walk or random-reactive controls. Budget experiments include matched static top-10 controls. This control design is load-bearing: two methods that look plausible in isolation are worse than their random controls.

5 Results

5.1 Channel sets drift at long reasoning horizons

Figure 1 shows the headline measurement. All four models exhibit increasing strict set-leaving across decode position. By the final measured position, strict set-leaving is 0.566 for

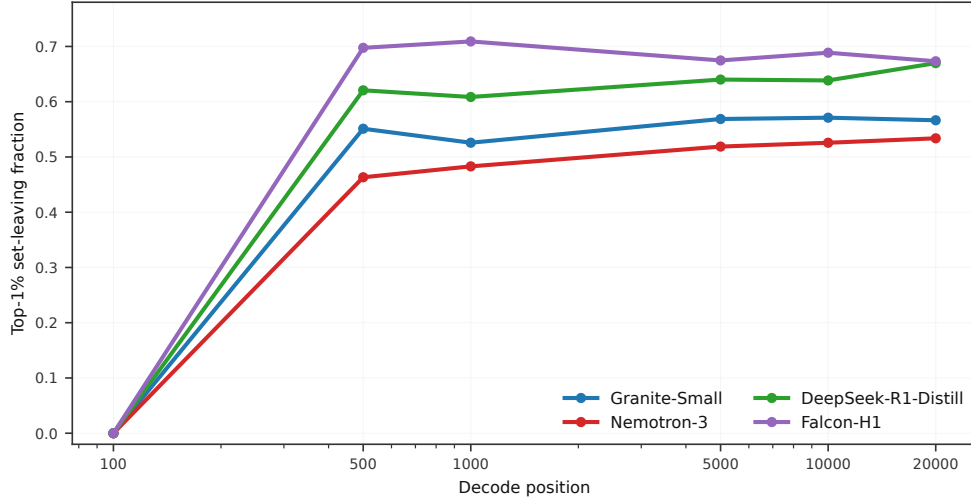


Figure 1: Strict top-1% channel-set leaving relative to decode position 100. Static channel protection loses 53–67% of its original protected set by the final measured position across four reasoning models.

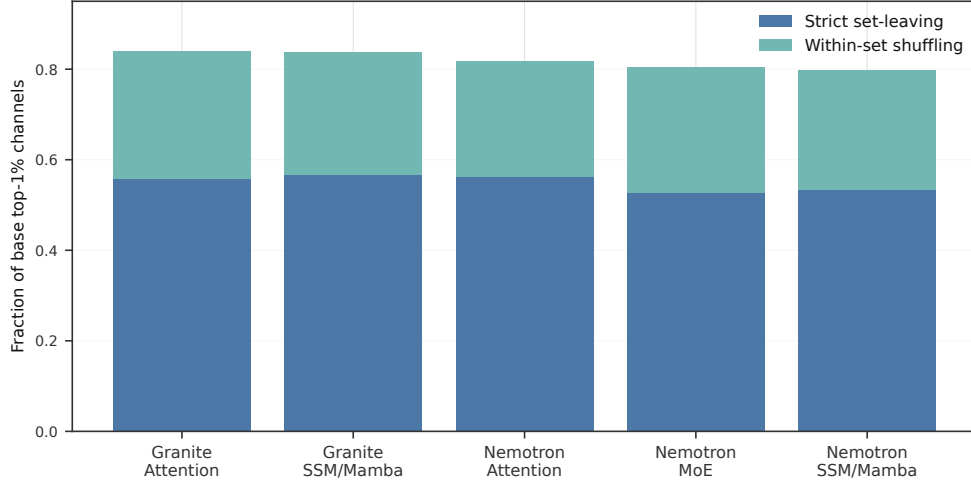


Figure 2: Layer-stratified drift at measured block outputs. Strict set-leaving is comparable across attention, SSM/Mamba, and MoE-classified layers, so the drift is not localized to one component type.

Granite-Small, 0.534 for Nemotron-3, 0.671 for DeepSeek-R1-Distill, and 0.674 for Falcon-H1. The pure-Transformer reference drifts at least as much as the hybrid models, so the phenomenon is not confined to Mamba layers.

5.2 Layer type does not explain the drift away

The Quamba2 comparison is strongest if drift appears in SSM/Mamba block outputs, not only in attention or MLP outputs. Figure 2 shows that Granite and Nemotron block-output drift is comparable across layer types. Granite attention and SSM/Mamba outputs have strict set-leaving of 0.557 and 0.567. Nemotron attention, MoE, and SSM/Mamba outputs have 0.563, 0.527, and 0.533. This does not isolate internal SSM state tensors, but it does show that the measured block-output surface is not channel-persistent.

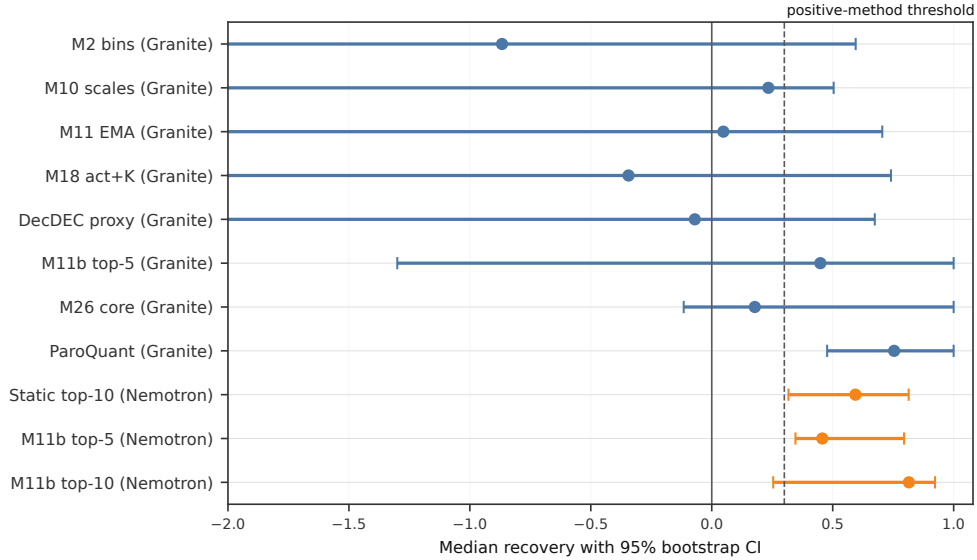


Figure 3: Median recovery with 95% bootstrap CIs. Long negative CI whiskers are clipped at the plot boundary for readability; exact rounded values are in Table 1. The dashed line marks the 0.30 positive-method threshold used in preregistered gates.

Method family	Model	Median recovery	95% CI	Key comparison
Position-conditioned channels	Granite	-0.867	[-3.44, 0.595]	random bin better by 0.668
Position-binned scales	Granite	0.234	[-2.07, 0.504]	random scale bin better by 0.761
EMA-smoothed top-1%	Granite	0.048	[-8.94, 0.705]	random-walk lower by 1.25
Activation+key coupling	Granite	-0.344	[-14.1, 0.741]	random coupled lower by 4.32
DecDEC proxy	Granite	-0.070	[-8.50, 0.674]	random reactive lower by 1.19
Stable core	Granite	0.178	[-0.116, 1.00]	random matched-core lower by 1.73
Budget EMA top-5	Granite	0.449	[-1.30, 1.00]	static top-10 lower by 0.501
Static top-10	Nemotron	0.594	[0.317, 0.814]	matched-budget control
Budget EMA top-5	Nemotron	0.457	[0.346, 0.795]	static top-10 higher by 0.138
Budget EMA top-10	Nemotron	0.815	[0.254, 0.923]	static top-10 lower by 0.220
ParoQuant rotation	Granite	0.754	[0.477, 1.00]	M11b top-5 lower by 0.304

Table 1: Intervention outcomes. Recovery is unclipped; values can be negative or exceed one. Values are rounded for the main text; CI endpoints displayed as 1.00 may slightly exceed one before rounding. ParoQuant is included as a rotation baseline and is not a channel-set method.

5.3 Channel-set interventions mostly fail at top-1% budget

Figure 3 and Table 1 summarize the intervention outcomes. Hard position-conditioned protection is outperformed by its random-bin control by 0.668 median recovery, and hard scale bins are outperformed by random scale-bin assignment by 0.761. EMA smoothing is different: M11 beats its random-walk control by 1.25 median, but its absolute recovery is only 0.048. M18 activation+key coupling has the strongest signal-vs-random margin among failed top-1% methods, beating random coupling by 4.32 median, but its absolute recovery is negative. DecDEC-style reactive selection is less bad than a random reactive control but still does not recover quality. M26 stable-core protection shows signal over random matched-core channels but remains below the positive-method threshold.

5.4 Budget is load-bearing, but not model-invariant

M11b changes the conclusion from uniformly negative to partially positive. On Granite-Small, top-5 EMA protection recovers 0.449 of the static gap and beats static top-10 by 0.501 median, but the CI is wide. On Nemotron, top-5 is positive with a strictly positive CI, yet it does not beat static top-10. The top-10 arm is the strongest channel-set result:

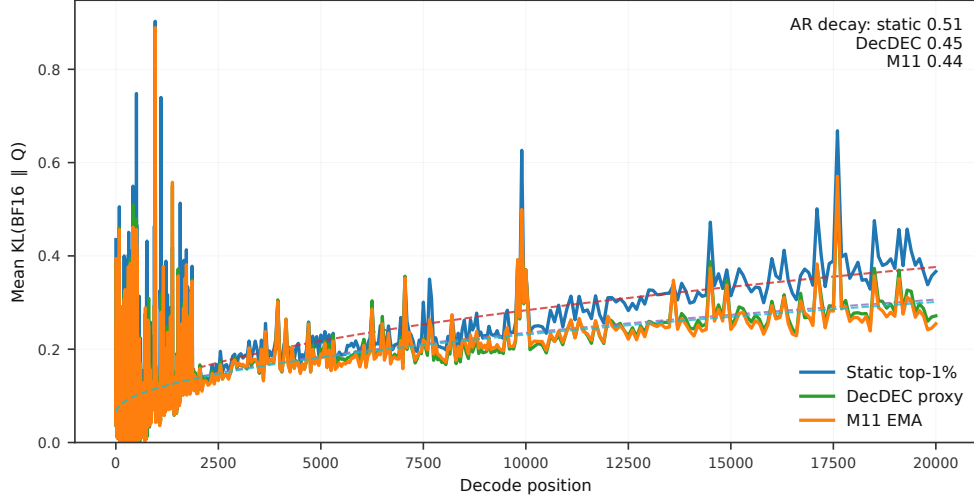


Figure 4: Per-position KL trajectories on Granite-Small. Among the tested functional forms, all three regimes have lowest residuals under sublinear square-root fits and moderate AR decay, rather than superlinear growth.

median recovery 0.815, CI [0.254, 0.923], and a 0.220 median margin over static top-10. Thus budget is a real mechanism, but the best budget and the value over static protection are model-dependent.

ParoQuant constrains the claim. Its Granite-Small rotation baseline recovers 0.754, exceeding M11b top-5 on the same model. The paper therefore does not argue that W4A16 PTQ is intrinsically broken; it argues that channel-set protection is fragile under long reasoning drift, while rotation-based conditioning can address part of the same endpoint by a different mechanism.

5.5 KL does not support runaway compounding

Figure 4 tests whether failed channel-set methods primarily fail because errors compound over decode. In the dense Granite-Small packet, static top-1%, DecDEC proxy, and M11 have lowest residuals under a sublinear square-root fit among the tested linear, sublinear, and superlinear alternatives. Mean KL is 0.150, 0.133, and 0.131 respectively, and fitted AR decays are 0.51, 0.45, and 0.44. This weakens compound error as the primary explanation in this measured regime and shifts attention back to stale or misbudgeted protection.

6 Discussion

The results support a three-mechanism account. First, boundary discontinuities are actively harmful: both hard protected-set and hard scale-bin switching lose to random controls. Second, smoothing removes the worst boundary harm but does not recover quality at top-1% budget; M11 and M18 contain signal, but not enough protected capacity. Third, budget is binding, but budget alone is not a universal recipe: Nemotron top-10 succeeds, Granite top-5 is promising but high-variance, and static top-10 is already strong on Nemotron.

Trace heterogeneity remains important. In the Granite channel-set experiments, roughly one third of the 12-trace slice has no recoverable static gap under the ratio metric; in the larger static-union experiment the no-gap fraction is 0.375. Early static-union gates retain no-gap traces as zero-recovery rows, while later method packets report recovery ratios on positive-gap traces and track the no-gap fraction separately. This limits median-based claims and explains the wide CIs on several methods. The paper therefore emphasizes controlled mechanism comparisons and cross-model directionality rather than a single universal method headline.

The natural next step is composition rather than another threshold-only channel set. ParoQuant shows that rotations can recover much of the Granite W4A16 gap; M11b shows that protected budget matters. A stronger ICLR follow-up would test rotation plus budget-aware protection, larger reasoning scales, and training-time or calibration-time interventions that make long-decode channel statistics less brittle.

7 Limitations

The scope is W4A16 PTQ on small reasoning models and measured block-output activation channels. We do not claim results for FP8, MXFP4/NVFP4, W4A4, or quantization-aware training. The hybrid-model Quamba2 comparison is at the accessible block-output surface, not at every internal SSM state tensor. Only Granite-Small has dense FFT and KL diagnostics; four-model dense spectral and per-position KL replication remain future work. The M11b positive result is cross-model but not model-invariant: Nemotron top-10 succeeds, while Granite’s strongest signal is top-5 with wide uncertainty.

8 Conclusion

Long-reasoning W4A16 channel protection fails when static protected sets drift, and our experiments show that this drift is substantial across four measured models. Budget-tuned smoothing partially remedies the failure, especially on Nemotron, but rotations and broader calibration changes remain the stronger path toward a deployable method.

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Ethics Statement

This work analyzes benchmark prompts and model activations. It does not use private user data or make deployment-safety claims. The main ethical risk is overclaiming a scoped mechanism study as a universal quantization recipe; the paper therefore separates measurement, mechanism, and partial-remedy claims.

Reproducibility Statement

The release package contains the public FAST_VERIFY interface, including release/README.md, release/docs/reproducing_results.md, and release/src/scripts/reproduce_all.sh. FAST_VERIFY replays frozen paper-claim values and validates the documented tolerance mapping. Full-fidelity GPU reruns are not implemented inside release/; they remain in the archived experimental packet runners and are disclosed as a release limitation.

Model	HuggingFace identifier	Snapshot commit
Granite-Small	ibm-granite/granite-4.0-h-small	b8c0982b
Nemotron-3	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16	cbd3fa9f
DeepSeek-R1-Distill	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	ad9f0ae0
Falcon-H1	tiiuae/Falcon-H1-0.5B-Instruct	8f2587ca

Table 2: Model checkpoints used by the experiments. Full provenance and file-level hashes are recorded in the archived result manifests; the release package records claim-to-script tolerance mappings.

Bootstrap seeds, deterministic trace indices, method budgets, and tolerance windows are specified in the release configs. GPU kernels can still exhibit minor nondeterminism across driver versions; the verification compares outputs to bootstrap CI tolerances rather than bit-exact scores. The original Qwen3.6 exploratory hook surface was not used for paper claims because activation access was incomplete.

LLM Usage Statement

The authors used LLM-assisted tools during project development for code drafting, paper-edit suggestions, figure-generation scripts, audit checklists, and summarization of local experimental artifacts. The authors remained responsible for experiment design, execution decisions, numerical verification, citation checking, and all final claims in this submission.

A Detailed Experimental Provenance

Paper description	Internal study label	Provenance location
Granite-Tiny pilot drift measurement	Phase 0	experimental/outlier_migrate/phase0/
Granite-Small drift replication	Phase 1	experimental/outlier_migrate/phase1/
Nemotron-3 drift replication	Phase 2	experimental/outlier_migrate/phase2/
Static migration-aware union, Granite-Tiny	Phase 3	experimental/outlier_migrate/phase3/
Static migration-aware union, Granite-Small	Phase 4	experimental/outlier_migrate/phase4/
DeepSeek Transformer control	Phase 5 prime	experimental/outlier_migrate/phase5_prime/
Falcon-H1 parallel-hybrid control	Phase 7	experimental/outlier_migrate/phase7/
Channel-set methods and baselines	Method study	experimental/outlier_migrate/phase9/
Paper release interface	Release package	release/docs/reproducing_results.md

Table 3: Translation from paper-facing experiment names to internal study labels. Exact packet directories, command lines, prompt hashes, and model file hashes are maintained in the release documentation and result manifests.

The valid Nemotron budget-tuning result uses the corrected static top-1% baseline rerun and reuses validated score caches for the budget arms. The earlier invalid baseline is excluded from all recovery claims. The release mapping table records both the valid rerun and the excluded failed attempt so that reviewers can audit the correction.

B Per-Method Detailed Results

Experiment	Included traces	No-gap fraction	Median recovery	95% CI
Static union, Granite-Tiny	24	0.417	0.000	[0.000, 0.711]
Static union, Granite-Small	24	0.375	0.000	[0.000, 0.0695]
Position-conditioned channels	8	0.333	-0.867	[-3.44, 0.595]
Position-binned scales	12	0.000	0.234	[-2.07, 0.504]
EMA-smoothed top-1%	8	0.333	0.048	[-8.94, 0.705]
Activation+key coupling	8	0.333	-0.344	[-14.1, 0.741]
DecDEC proxy	8	0.333	-0.070	[-8.50, 0.674]
Budget EMA top-5, Granite	8	0.333	0.449	[-1.30, 1.00]
Static top-10, Nemotron	10	0.167	0.594	[0.317, 0.814]
Budget EMA top-5, Nemotron	10	0.167	0.457	[0.346, 0.795]
Budget EMA top-10, Nemotron	10	0.167	0.815	[0.254, 0.923]
Stable-core protection	8	0.333	0.178	[-0.116, 1.00]
ParoQuant rotation	8	0.333	0.754	[0.477, 1.00]

Table 4: Detailed rounded outcomes for all method families discussed in the main text. Values are recovery ratios relative to each experiment’s BF16-vs-static baseline.

C Extended Layer-Stratified Results

Model	Layer type	Layers	Strict set-leaving	Within-set shuffling
Granite-Tiny	Attention	4	0.618	0.186
Granite-Tiny	SSM/Mamba	36	0.636	0.174
Granite-Small	Attention	4	0.557	0.282
Granite-Small	SSM/Mamba	36	0.567	0.270
Nemotron-3	Attention	6	0.563	0.255
Nemotron-3	MoE	23	0.527	0.278
Nemotron-3	SSM/Mamba	23	0.533	0.264

Table 5: Layer-type decomposition. Strict set-leaving is the component most directly relevant to protected-set membership; within-set shuffling motivates scale-refresh mechanisms but is not itself membership loss.